

Quantum-Classical Equivalence for AND Functions

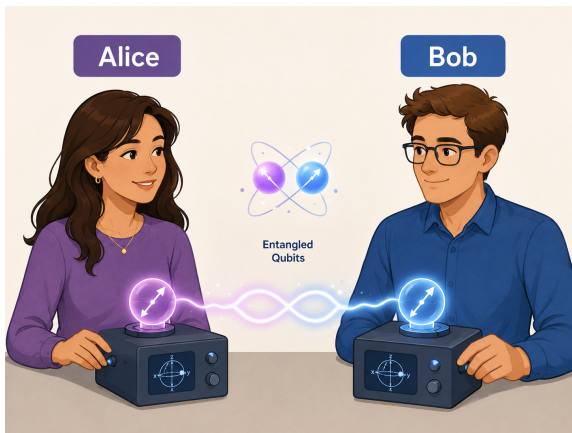
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introducing quantum communication

Our model of quantum communication: Alice and Bob share an unlimited number of EPR pairs; have their own auxiliary quantum systems, and can communicate classical bits ¹. Cost of the communication protocol is the number of classical bits communicated.



Quantum-Equivalence Conjecture: For a total function $f : \{0, 1\}^n \rightarrow \{0, 1\}$,

$$R(f) \leq \text{poly}(Q(f))$$

where

- $R(f)$ = classical randomized communication complexity
- $Q(f)$ = quantum communication complexity (with prior entanglement)

Major open problem in communication complexity. First formally proposed by Shi and Zhu (2008)

Our result: For AND functions, $f = g \circ \text{AND}_2$,

$$D(f) \leq \text{poly}(Q(f), \log(n))$$

where $D(f)$ = classical deterministic communication complexity.
Thus, Quantum Equivalence Conjecture is true for AND functions up to $\text{polylog}(n)$ factors.

This answers an open problem of Razborov [2002].

Why study AND functions?

- Lifted functions form a large class of interesting communication problems. They form a testbed for all major conjectures in communication complexity (e.g, Log Rank Conjecture, Log Approximate Rank Conjecture [now known to be false], Quantum Equivalence Conjecture)

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- Of particular interest is the case when the lifting gadget is as simple as possible, i.e., it has only two bits. In this case, the gadget does not have too much ability to obfuscate the inputs between Alice and Bob.
- There are only two nontrivial gadgets on 2 bits: XOR and AND. Both XOR and AND functions have been extensively studied in the literature.

AND functions

- There is a base function $g : \{0, 1\}^n \rightarrow \{0, 1\}$. Alice has an input $x \in \{0, 1\}^n$, Bob has an input $y \in \{0, 1\}^n$. Their goal is to compute

$$g(x_1 \wedge y_1, \dots, x_n \wedge y_n)$$

Prior work

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- **Kremer (1996)** proved log-approximate rank lower bounds quantum communication complexity.
- **Lovett, Knop, McGuire, Yuan (2021)** proved the log rank conjecture for AND functions. By an intermediate result in their proof, it became clear that proving the relation conjectured by Burhman and deWolf would also imply the Quantum Equivalence Conjecture for AND functions.

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Prior work

- **Chattopadhyay, Dahiya, Lovett (2026)** proved that de-Morgan sparsity and approximate de-Morgan sparsity are polynomially related.
- They also show the Quantum Equivalence Conjecture for functions of the form $g \circ \text{EQ}_4$.
- We extend their work to show the relation between quantities conjectured by Burhman and deWolf. This implies the Quantum-Equivalence Conjecture for AND functions.

prior results

- A result of Kremer ²:

$$Q^{cc}(F) \geq \text{poly}(\log(\text{rank}^{1/3}(F)))$$

where

$$\text{rank}^{1/3}(F) = \inf\{\text{rank}(G) \mid \|G - F\|_{\infty} \leq 1/3\}$$

²The result of Kremer only lower bounds clean-quantum communication complexity.[BdW01] observed that it readily implies general quantum communication lower bounds too. Another proof was given by Lee and Shraibman [LS07]

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- Thus, log-approximate rank conjecture for a class of functions also implies quantum-equivalence conjecture for that class of functions.
- It is known that LARC is false in general (**Mande, Suhail, Chattopadhyay, 2019**). Nevertheless, in this work we show LARC is true for AND functions (upto $\text{polylog}(n)$ factors); thereby also establishing the QEC for AND functions.

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prior results

In case Yogesh has defined this properly in his previous talk I shall skip the next few slides.

- Any function $g : \{0, 1\}^n \rightarrow \mathbb{R}$ has a unique representation as a multilinear polynomial:

$$g(x) = \sum_{S \subseteq [n]} \alpha_S x_S$$

where $x_S = \prod_{i \in S} x_i$.

Note: This is not the Fourier decomposition where the domain is $\{\pm 1\}$. Also note that we are working over $\mathbb{R}$, not $\mathbb{F}_2$.

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Define sparsity of g , $\text{spar}(g)$, to be the number of nonzero monomials in this representation.

- The conjecture in [BdW01]:

$$\text{rank}^{1/3}[g \circ \text{AND}_2] \approx \text{spar}(g)$$

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- The conjecture in [BdW01]:

$$\text{rank}^{1/3}[g \circ \text{AND}_2] \approx \text{spar}(g)$$

- Our main result:

$$\log \text{spar}(g) \leq \text{poly} \left(\log \left(\text{rank}^{1/3}(g \circ \text{AND}_2), \log(n) \right) \right)$$

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How this implies the Quantum Equivalence Conjecture

- Recall we want to upper bound $D(f)$ in terms of $\log \left(\text{rank}^{1/3}(f \circ \text{AND}_2) \right)$

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- **Knop, Lovett, McGuire, Yuan** showed that

$$\log D(g \circ \text{AND}_2) \leq \text{poly}(\log \text{spar}(g), \log(n))$$

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- **Knop, Lovett, McGuire, Yuan** showed that

$$\log D(g \circ \text{AND}_2) \leq \text{poly}(\log \text{spar}(g), \log(n))$$

- Since our result upper bounds $\text{spar}(g)$ in terms of $\text{rank}^{1/3}(g \circ \text{AND}_2)$, the result immediately follows.

Goal: Upper bound $\text{spar}(g)$ in terms of $\text{rank}^{1/3}(g \circ \text{AND}_2)$

- Assume, for the sake of contradiction, $\text{spar}(g)$ is large and $\text{rank}^{1/3}(g \circ \text{AND}_2)$ is small.

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- Assume, for the sake of contradiction, $\text{spar}(g)$ is large and $\text{rank}^{1/3}(g \circ \text{AND}_2)$ is small.
- From the second assumption, we have functions

$$a_i, b_i : \{0, 1\}^n \rightarrow [0, 1], i = 1, 2, \dots, m$$

and constants $\alpha_i, i = 1, 2, \dots, m$ such that

$$\left| g(x \wedge y) - \sum_{i=1}^m \alpha_i a_i(x) \wedge b_i(y) \right| \leq 1/3$$

- By standard techniques (eg John's lemma), we can assume $\sum |\alpha_i| \leq L$ (where $\log L \leq \text{polylog}(m)$)

Goal: Upper bound $\text{spar}(g)$ in terms of $\text{rank}^{1/3}(g \circ \text{AND}_2)$

- We shall construct a function f (to be obtained from g) such that
 - Approximate degree of f is small
 - Degree of f is high

This contradicts known relations between degree and approximate degree (first proven by Nisan-Szegedy; constants were optimized later).

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This contradicts known relations between degree and approximate degree (first proven by Nisan-Szegedy; constants were optimized later).

- We shall get a low-degree approximating polynomial from the inequality

$$\left| g(x \wedge y) - \sum_{i=1}^m \alpha_i a_i(x) \wedge b_i(y) \right| \leq 1/3$$

whereas the high sparsity of g will guarantee the original polynomial has high degree.

Goal: Upper bound $\text{spar}(g)$ in terms of $\text{rank}^{1/3}(g \circ \text{AND}_2)$

- Note that every function $f : \{0, 1\}^n \rightarrow \mathbb{R}$ can also be represented as a function $\tilde{f} : \{\pm 1\}^n \rightarrow \mathbb{R}$. It is easy to see that their degrees are the same.

Convention: For a function $f : \{0, 1\}^n \rightarrow \mathbb{R}$, we denote its equivalent function with the domain $\{\pm 1\}^n$ by $\tilde{f} : \{\pm 1\}^n \rightarrow \mathbb{R}$.

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- For upper-bounding the approximate degree, we shall look at functions in the $\{\pm 1\}^n$ domain. For lower bounding the degree, we shall look at functions in the $\{0, 1\}^n$ domain. Since the degree is invariant under change of domain, this is ok.

- **Observation:** Given the expression

$$g(x \wedge y) \approx \sum_{i=1}^m \alpha_i a_i(x) \wedge b_i(y),$$

we can hit each a_i by a restriction+averaging operator (ρ, δ) , replace each b_i by a constant, and get an approximation for g_ρ

Restriction+averaging operators:

A **restriction+averaging operator** is denoted by a string

$R \in \{0, 1, *, \Delta\}^n$. It converts a function $f : \{0, 1\}^n \rightarrow \mathbb{R}$ to a function $f : \{0, 1\}^{\text{free}(R)} \rightarrow \mathbb{R}$ where $\text{free}(R) = \{i \mid R_i = *\}$. The transformation is as follows:

- If $R_i = b \in \{0, 1\}$, fix the i -th coordinate to b .
- If $R_i = \Delta$, average over the i -th coordinate.
- If $R_i = *$, keep the i -th coordinate free.

A partial assignment $\rho \in \{0, 1, *\}^n$ is said to be consistent with R if it can be obtained from R in the following manner:

- Change some of the 1's to 0's.
- Change all of the Δ 's to 0.

Goal: Upper bound $\text{spar}(g)$ in terms of $\text{rank}^{1/3}(g \circ \text{AND}_2)$

- We shall define a distribution $\mathcal{D}$ over (R, ρ) (ρ consistent with R) such that for any $\tilde{a} : \{\pm 1\}^n \rightarrow [0, 1]$,

$$\mathbb{E}_{(R, \rho) \sim \mathcal{D}}[|\tilde{a}_R|^{\geq d}] \leq \varepsilon$$

where $|\tilde{f}|^{\geq d}$ denotes ℓ^1 Fourier weight above level d

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- ... and such that $\mathbb{E}_{(R, \rho) \sim \mathcal{D}}[\deg(f_\rho)] \geq D$

- The first property, along with the expression for approximate rank, will give that $\text{approx-deg}(g_\rho) \leq d$ for a typical $R \sim \mathcal{D}$

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- The second property will give that $\text{deg}(g_\rho) \geq D$ for a typical $R \sim \mathcal{D}$.
- If the gap between d and D is large enough, we are done.

Obtaining an approximation for f_ρ

- Recall the expression we got from the low-approximate rank assumption:

$$g(x \wedge y) \approx \sum_{i=1}^m \alpha_i a_i(x) b_i(y)$$

$$a_i, b_i : \{0, 1\}^n \rightarrow [0, 1]$$

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- Claim:** Given a restriction+averaging operator R , with ρ its associated partial assignment, there exist constants $c_1, c_2, \dots, c_m \in [0, 1]$ such that

$$g_\rho(z) \approx \sum_{i=1}^m \alpha_i c_i(a_i)_R(z)$$

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- I will show this only for two special cases. The general case should be clear from this.

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- Plugging $y_0 = (0, 1, 1, \dots, 1), x = (1, z_2, z_3, \dots, z_n),$ we get

$$g(0, z_2, z_3, \dots, z_n) \approx \sum_{i=1}^m \alpha_i a_i(1, z_2, \dots, z_n) b_i(y_0)$$

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- Average these two expressions:

$$g(0, z_2, \dots, z_n) \approx \sum_{i=1}^m \alpha_i b_i(y_0) (a_i)_R(z)$$

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- **Special case 2:** $R = \{1, *, *, *, \dots, *\}, \rho = \{0, *, *, *, \dots, *\}.$

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This is the desired expression.

Exhibiting a gap between degree and approximate degree

- **Conclusion:** If R is a restriction+averaging operator and ρ is a consistent partial assignment, we get an approximation for g_ρ in terms of the $(a_i)_R$'s.

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- **Conclusion:** If R is a restriction+averaging operator and ρ is a consistent partial assignment, we get an approximation for g_ρ in terms of the $(a_i)_R$'s.
- **Our next goal:** Using the fact that sparsity of g is high, we define a distribution $\mathcal{D}$ over (R, ρ) such that:
 - ρ is consistent with R
 - For all $a : \{0, 1\}^n \rightarrow [0, 1]$, $E_{R, \rho \sim \mathcal{D}}[\tilde{\deg}(a_R)] \leq d$ (we will do something slightly different actually)
 - $E_{(R, \rho \sim \mathcal{D})}[\deg(g_\rho)] \geq D$

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 - $E_{(R, \rho) \sim \mathcal{D}}[\deg(g_\rho)] \geq D$
- If we can do this with a sufficiently big gap between d and D , from the previous observation we get that the approximate degree of g_ρ is very small; whereas the second property implies the exact degree of g_ρ is very large. But Nisan-Szegedy implies for any Boolean function, degree and approximate degree are polynomially related. Contradiction.

Simplifying any bounded function

First requirement:

- This is what we will show:
For any $\tilde{a} : \{\pm 1\}^n \rightarrow [0, 1]$,

$$\mathbb{E}_{R, \rho \sim \mathcal{D}}[||\tilde{a}_R||^{\geq d}] \leq \varepsilon$$

where $||f||^{\geq d}$ denotes total ℓ^1 Fourier mass above level d .
Notice that we switched to $\{\pm 1\}$ domain here.

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- If we achieve this, the total ℓ^1 Fourier mass above level d of the function

$$\sum_{i=1}^L \alpha_i c_i(a_i)_R$$

will be at most $L\varepsilon$, which we will set to be $< 1/100$. Thus, on expectation, the approximate degree of g_ρ will be $\leq d$.

Simplifying any bounded function

Why switching to the $\{\pm 1\}$ basis is helpful:

- Let $\tilde{f} : \{\pm 1\}^n \rightarrow \mathbb{R}$ be any function. Its Fourier expansion is

$$\tilde{f} = \sum_{S \subseteq [n]} \hat{\tilde{f}}(S) \chi_S$$

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- Suppose we hit $\tilde{f}$ by a restriction+averaging operator R , which averages over some subset T of variables. Then, we have

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- Thus, averaging out over a variable kills all monomials containing it.

Defining the distribution

- Assume $g : \{0, 1\}^n \rightarrow \{0, 1\}$ has $\geq s$ non-zero monomials in the deMorgan basis. Let $\mathcal{F} \subseteq 2^{[n]}$ denote the set of nonzero monomials.

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- Let $T \subseteq [n]$ be a shattered set of size $\geq \Omega(\log(s)/\log(n))$.
- For any $A \subseteq T$, there exists $S \in \mathcal{F}$ such that $S \cap T = A$.

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- Let $T \subseteq [n]$ be a shattered set of size $\geq \Omega(\log(s)/\log(n))$.
- For any $A \subseteq T$, there exists $S \in \mathcal{F}$ such that $S \cap T = A$.

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- For every $i \in T$, set

$$R(i) = \begin{cases} * & \text{with probability } p \\ \Delta & \text{with probability } 1 - p \end{cases}$$

Set $R(i) = 1$ for all $i \notin T$.
(Here p is small but non-negligible.)

Defining the distribution

- Let $A = \{i | i \in T, R(i) = *\}$. Set

$$\rho(i) = \begin{cases} * & \text{for } i \in A \\ 0 & \text{for } i \in T \setminus A \end{cases}$$

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- Look at the polynomial

$$g_{A,T} = \sum_{S: S \cap T = A} \alpha_S x_{S \setminus A}$$

This is defined on $\{0, 1\}^{[n] \setminus T}$, and is nonzero since T is shattered.
 So there exists a $y \in \{0, 1\}^{[n] \setminus T}$ such that $g_{A,T}(y) \neq 0$.

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- Set $\rho(i) = y_i$ for $i \notin T$.

this distribution is good

Recall that we have to show the following properties about this distribution:

- 1 **It simplifies any bounded function:** For any $\tilde{f} : \{\pm 1\}^n \rightarrow [0, 1]$,

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- 2 **It preserves high degree of g :**

$$\mathbb{E}[\deg(g_\rho)] \geq d$$

Preserving high degree

Claim: $E[\deg(g_\rho)] \geq p|T|$

Proof.

The coefficient of x_A in g_ρ is nonzero, and we have $E[|A|] = p|T|$. We chose $y \in \{0, 1\}^{[n] \setminus T}$ such that $g_{A,T}(y) = \sum_{S \cap T = A} \alpha_S y_{S \setminus A} \neq 0$. This is precisely the coefficient of the monomial x_A in g_ρ . □

Destroying all bounded functions

Claim: For all $\tilde{f} : \{\pm 1\}^n \rightarrow [0, 1]$,

$$\mathbb{E}[\|\hat{\tilde{f}}_R\| \geq d] \leq \varepsilon$$

Proof.

Recall how R is sampled.

- Set everything outside T to 1.
- Set everything in T to $*$ with probability p , Δ with probability $1 - p$.



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- So we can first think of hitting f with the fixed restriction δ that fixes everything outside T to 1. Letting $h = f_\delta$, we set every variable of h to $*$ with probability p ; otherwise we set it to Δ .



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- If a monomial contains any variable that gets hit by Δ , that monomial dies.
- The probability of a monomial $\hat{h}(S)_{\chi_S}$ surviving, therefore, is $p^{|S|}$. For large monomials this is small.



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Expected Fourier mass above level $d \leq \sum_{|S| \geq d} p^{|S|} |\hat{h}(S)|$



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- Break this sum by layers $k = d, d+1, \dots, n$. At layer k , the sum is

$$p^k \sum_{|S|=k} |\hat{h}(S)|$$



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- At layer k , the sum is

$$p^k \sum_{|S|=k} |\hat{\tilde{h}}(S)|$$

- Since $\sum_S \hat{\tilde{h}}(S)^2 \leq 1$, by Cauchy-Schwartz, the sum above is

$$\leq p^k \sqrt{\binom{n}{k}} \leq \left(p \sqrt{\frac{en}{k}}\right)^k$$



Destroying all bounded functions

Claim: For all $\tilde{f} : \{\pm 1\}^n \rightarrow [0, 1]$,

$$\mathbb{E}[\|\hat{\tilde{f}}_R\|^{\geq d}] \leq \varepsilon$$

Proof.

- Note that $k \geq d$. Choose $d \approx n^{1/100}$, $p \approx d^{-99/200-\epsilon}$. Win.



Conclusion

- Log-approximate rank conjecture is true for AND functions.
Therefore, Quantum-Equivalence conjecture is also true.
- **Open:** What about quantum-equivalence conjecture for XOR functions? It is known that LARC is false for XOR functions, so we need some other technique to lower bound quantum communication complexity.
- **More ambitious:** What about quantum-equivalence conjecture for *all* total functions? This would be a very major breakthrough.